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Assignment: Artificial Intelligence

Title:

“Personalized Restaurant Recommendation Agent (Rule-Based AI Agent)”

1. Problem Identification in Terms of an Agent

Diners today have countless restaurants to choose from, but not every restaurant suits their current occasion, budget, or dietary needs, and manually comparing menus, prices, and reviews for every option takes time. This problem fits an intelligent agent because it needs a system that can sense the diner's details and a candidate restaurant (its environment), reason over this information using a set of predefined preference rules, and act by producing a recommendation that helps the diner decide whether to visit. The agent designed here, called the Personalized Restaurant Recommendation Agent, is a rule-based agent: it does not learn from data, but works with a fixed knowledge base of IF-THEN rules built around cuisine preference, occasion, budget, and dietary restriction, checking a diner's profile and a candidate restaurant against these rules to produce a simple, personalized recommendation with reasoning.

2. Problem Formulation

Input: Diner's age, favorite cuisine, current occasion, budget range, dietary restriction, and the candidate restaurant.

Dataset: A set of predefined recommendation rules built from cuisine match, occasion suitability, budget, and dietary needs, i.e.,

Age	Favorite Cuisine	Occasion	Budget	Candidate Restaurant	Rule / Condition	Recommendation
25	Italian	Date Night	Mid-range	La Bella Notte	Cuisine and occasion both match	Highly Recommended
34	Chinese	Casual Outing	Low	Golden Dragon	Cuisine matches, budget fits	Recommended
50	Seafood	Business Meeting	High	Ocean Grill	Occasion does not match cuisine style	Maybe
19	Fast Food	Family Outing	Low	Burger Hub	Cuisine and budget match	Recommended
40	Vegan	Health-Conscious Meal	Mid-range	Green Bowl	Strong cuisine and dietary match	Highly Recommended

Algorithm: Rule-Based AI using IF-THEN matching, where the diner's input is compared against the rules in the dataset to reach a decision.

Output: Highly Recommended / Recommended / Maybe / Not Recommended, along with a short reason.



3. Evaluation in the Context of PEAS

PEAS Element	Description
Performance Measure	How relevant the restaurant match is to the diner, overall diner satisfaction, correctness of matching cuisine, occasion and budget, and how quickly the suggestion is produced.
Environment	The diner's personal profile (age, favorite cuisine, occasion, budget, dietary restriction), the restaurant database, and the interface through which the diner enters information and views results.
Actuators	The output/display module that shows the recommendation label (Highly Recommended / Recommended / Maybe / Not Recommended) along with a short reason.
Sensors	The input module that reads the diner's entered or stored details: age, favorite cuisine, occasion, budget, dietary restriction, and the restaurant being queried.

4. Environment Properties

Observability: Partially observable

The agent can only work with the details a diner chooses to give it, such as favorite cuisine, occasion, and budget. It has no way of sensing things the diner has not entered, like their true craving at that exact moment or restaurants they have tried elsewhere without recording them, so its view of the diner is incomplete.

Determinism: Deterministic within the rule base

Since the agent is rule-based, giving it the same diner profile and the same candidate restaurant will always trigger the same rules and produce the same recommendation. There is no randomness involved in how it reaches its decision.

Episodic vs. Sequential: Episodic

Each restaurant query is handled as its own separate episode. The recommendation given for one restaurant does not depend on what was recommended earlier for a different restaurant, even though the diner's saved profile stays the same across queries.

Static vs. Dynamic: Semi-static (mostly static)

While the agent is working out a single recommendation, the diner's profile and the restaurant database stay fixed. Changes only happen between sessions, for example when the diner updates their budget or dietary need, so the environment is treated as static during each individual query.

Discrete vs. Continuous: Discrete

Cuisines, occasions, budget tiers, and recommendation levels are all represented as a limited set of fixed categories rather than continuous values, and the rules are written to match these discrete categories.

Known vs. Unknown: Known

The rules connecting cuisine, occasion, budget, and dietary need to a recommendation are decided in advance by the designer based on general dining patterns, so the agent already knows how its environment behaves before it starts working.

Single-agent vs. Multi-agent: Single-agent

The system works as a single personal recommendation assistant for one diner at a time. It does not need to cooperate or compete with any other independent agent while making its decision.